Toric and Semitoric Packing Capacities

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Introduction and Examples

Goals

- Understand properties of packing capacity for Delzant and semitoric polygons.
- Develop an algorithm to find the maximum valid packing of a Delzant polygon.

Definition. *A **Delzant Polygon** is a rational convex polygon such that all vertices satisfy the **Delzant Condition**, namely that the two primitive direction vectors $\vec{u}, \vec{v}$ of the adjacent edges satisfy* $\det(\vec{u}, \vec{v}) = \pm 1$.

Definition. *A **Semitoric Polygon** is a rational convex polygon with **marked points** and **cuts** (a line segment from a marked point to its vertex) such that each vertex satisfies one of the following:*

(I) The vertex is on a cut and for $\vec{v}, \vec{w}$ primitive edge direction vectors in the counter-clockwise direction, either $\det(\vec{v}, T\vec{w}) = 0$ or $\det(\vec{v}, T\vec{w}) = \pm 1$, where

$$T = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

*The former is called the **fake corner condition** while the latter is called the **hidden corner condition**.*

(II) The vertex is not on a cut and it satisfies the Delzant condition.

Cuts can be up or down, and their direction can be changed by applying T (up to down) or T^{-1} (down to up) appropriately to the portion of the polygon right of the cut.

Definition. *A **Valid Packing** of a Delzant or semitoric polygon is a set of triangles placed at the vertices such that:*

(I) Each triangle is an image under some $A \in \mathrm{SL}_2(\mathbb{Z})$ of the right isosceles triangle with $x \geq 0, y \geq 0$, and $x + y \leq \lambda$.

(II) The triangles are non-intersecting (but they can share vertices).

(III) If the polygon is semitoric, no triangle is packed at a fake corner, and no triangle contains a marked point.

*The **Packing Capacity** is defined as the ratio of the maximum valid packing (defined as a valid packing with the maximum total triangle area) to the area of the polygon* [1,2].

Definition. *The $\mathrm{SL}_2(\mathbb{Z})$ **Length** of a vector is defined as the real length of the vector after transformation under some $A \in \mathrm{SL}_2(\mathbb{Z})$ to either the x-axis or y-axis (the length is the same either way).*

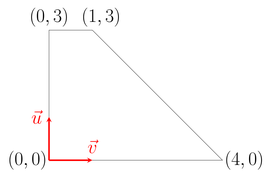


Figure 1: *Example of a Delzant polygon, with primitive edge direction vectors $\vec{u}$ and $\vec{v}$ at vertex $(0, 0)$.*

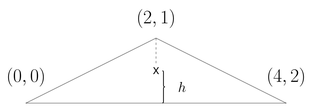


Figure 2: *Example of a semitoric polygon. x is the marked point, the dashed line is the cut (which is up), and h is the vertical distance of the marked point from the opposite side.*

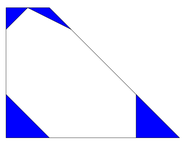


Figure 3: *Valid packing of the Delzant polygon in figure 1.*

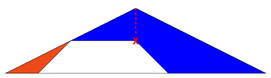


Figure 4: *Valid packing of the semitoric polygon in figure 2. Note that the blue "triangle" is deformed since it is packed at a marked corner. The cut and marked point are colored red for visibility.*

Methods, Results, and Algorithm

Methods:

Our goal is to find the maximum packing of the polygon, which amounts to maximizing our objective function $f(\mathbf{\Lambda}) = \frac{1}{2}(\lambda_1^2 + \ldots + \lambda_n^2)$ given the constraints $\lambda_i + \lambda_{i+1} \leq \ell_i$, where the ℓ_i are the $\mathrm{SL}_2(\mathbb{Z})$-lengths of the edges of the polygon. These constraints determine a polytope $\Delta_\lambda \subseteq \mathbb{R}^n$.

Theorem 1. *$f(\mathbf{\Lambda})$ restricted to Δ_λ obtains maxima only on the vertices of Δ_λ, where a vertex is defined as a 0-dimensional polytope residing at the intersection of $n-1$ facets of of Δ_λ.*

Theorem 1 uses the positive-definite Hessian of the objective function along with a change of basis provided by orthogonal vectors in affine geometry to apply the Extreme Value Theorem. This result allows packing capacity to be solved in $\mathcal{O}\binom{2n}{n}$ time complexity.

Definition. *We call the constraint $\ell_i \leq \ell_{i+1}$ and $\ell_i \leq \ell_{i-1}$ the **forward and backward alternating edge conditions**, respectively. An alternating edge condition is said to be **fully satisfied** on a subset of edges if each edge in the subset satisfies the forward condition or each edge satisfy the backward condition.*

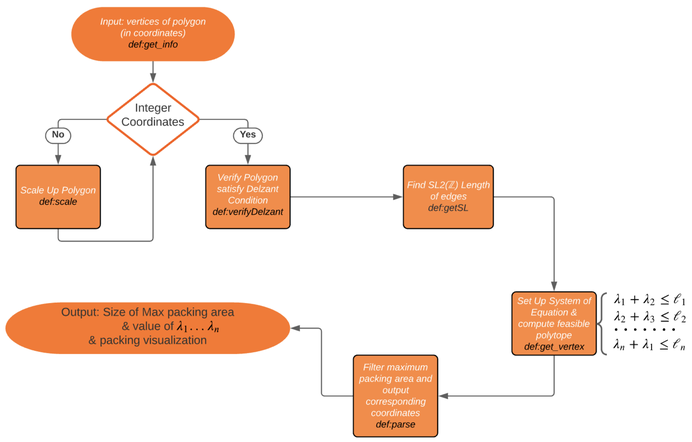
Theorem 2. *Given a $n = 2m$ dimensional Δ_λ, any $\mathbf{\Lambda} \in \Delta_\lambda$ has a maximum packing area given by*

$$\max_{\Delta_\lambda} \frac{1}{2}||\mathbf{\Lambda}||_2^2 = \frac{1}{2} \sum_{i \text{ odd}}^{n} \ell_i^2$$

if and only if one of the alternating edge conditions is fully satisfied for all odd edges.

Theorem 2 uses the fact that the 2-norm is bounded above by the 1-norm to provide an achievable upper bound for our packing area. This result reduces the time complexity to $\mathcal{O}(n)$ time if the alternating edge condition is satisfied.

Algorithm:



Function Logic:

def:Scale(vertices)
Iteration in vertices and get lcm of coordiate demoninators. Then multiply lcm to each coordinate in vertices.

def:show_graph(Vertices, sl2_length, set, tittle="")
Draw the original polygon as background. Determine the coordinates of packing area by ratio of λ to SL2($\mathbb{Z}$)

def:verifyDelzant(vert, nA, nB)
Compute two lines in vector form. Then check whether the determinant of composed matrix has absolute value of 1.

def:getSL(vert, nvert)
Take gcd of vert and nvert in vertical difference and horizontal difference

def:get_edges(vertices)
Get the list of all SL2($\mathbb{Z}$) in the graph by calling getSL

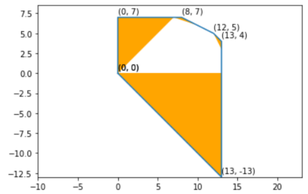
def:get_candidate(SL_length, k)
Suppose n elements in SL_length. Take k inequalities in equations system, then fullfill (n-k) rows with $\lambda_i = 0$. Find valid polygon from calculation in iteration.

def:get_vertex(SL_length)
Call get_candidate function with possible value of k, then sort all possible packing with increasing order of their magnitude.

def:parse(SL_length)
Find all qualified packings which share the same maximum packing magnitude.

def:show_info(Vertices, all=False)
Call all previous functions and compute all qualified packings. Output exhaustive information display as well as packing visualization. Notice: if input all = True. This function would output every possible maximum packing. Otherwise, it outputs a random maximum packing.

Demo Example. The 2D polygon with vertices: (0,0); (13,-13); (13,4); (12,5); (8,7); (0,7).
Output of program: This is the array of $\mathrm{SL}_2(\mathbb{Z})$
Length: [13. 17. 1. 2. 8. 7.]
The following picture shows the packing vertices and their magnitude: Vertex Coordinate: (0.0, 13.0, 1.0, 0.0, 1.0, 7.0) Magnitude: 110.0



Future Directions

Semitoric Polygons The semitoric case can be viewed as an expansion of the Delzant case. Future directions for this class of polygon include:

- Discover any interesting patterns or methods pertaining to the packing of semitoric polygons
- Adapting the current algorithm to compute the maximum valid packings of semitoric polygons

Delzant Polygons There is still much more work that can be done in the Delzant case. Future directions include:

- Determine if there is an analogous result to the even alternating edge case for Delzant polygons with an odd number of edges
- Further study of the behavior of corner chops on the packings of Delzant polygons
- Add more features to the program: Interactive grapher where a user can draw an input polygon, handling of irrational vertices, performing corner chops on the input polygon, and implementing new technologies to decrease time complexity.

References

[1] A. Figalli, J. Palmer, and Á. Pelayo. Symplectic G-capacities and integrable systems. *Ann. Sc. Norm. Super. Pisa Cl. Sci. (5)*, 18(1):65â103, 2018.

[2] Á Pelayo, Toric symplectic ball packing, *Topology and its Appl. 157* (2006), 3633â3644.

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